

Major Project 6 Sem

Design and Evaluation of an End-to-End Breach Lifecycle Simulation in Containerized Environments

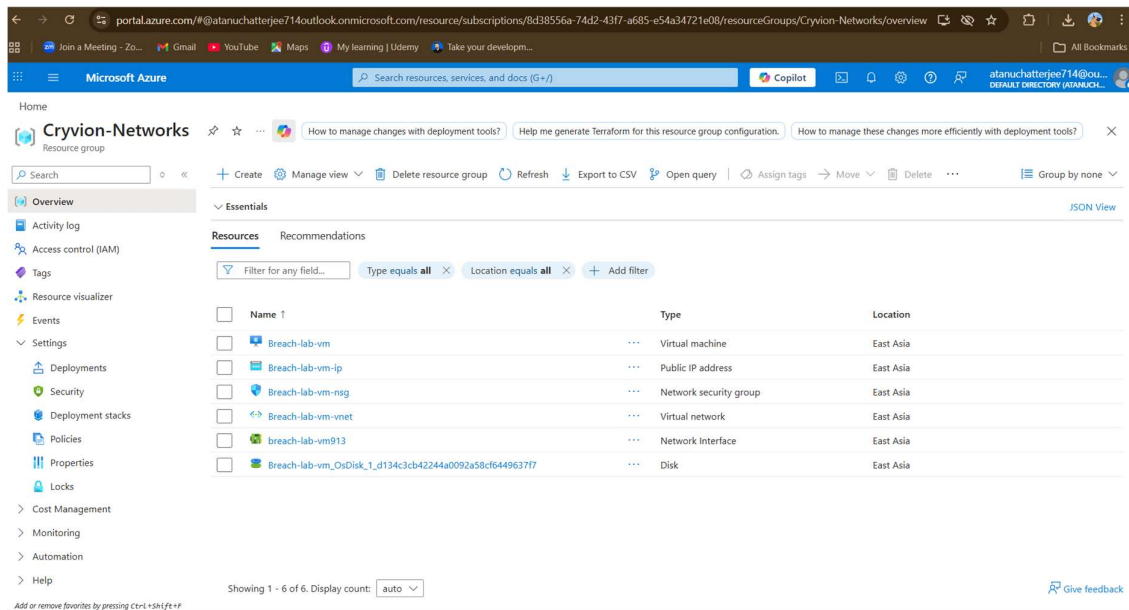
1. Abstract

This project demonstrates a real-world enterprise breach simulation using Microsoft Azure cloud infrastructure. A vulnerable web application (DVWA) was deployed using Docker, and multiple attacks such as network scanning, SQL injection, and XSS were performed.

Wazuh SIEM was used to detect and analyze these attacks.

2. Azure Environment Setup

Screenshot:



Name	Type	Location
Breach-lab-vm	Virtual machine	East Asia
Breach-lab-vm-ip	Public IP address	East Asia
Breach-lab-vm-nsg	Network security group	East Asia
Breach-lab-vm-vnet	Virtual network	East Asia
Breach-lab-vm913	Network interface	East Asia
Breach-lab-vm_OsDisk_1_d134c3cb42244a0092a58cf6449637f7	Disk	East Asia

Explanation:

- VM Name: Breach-lab-vm
- OS: Ubuntu 24.04
- Public IP: 20.189.64.115
- Region: East Asia

- Resource Group: Cryvion-Networks

3. Network Configuration

Screenshot:

The screenshot displays the Azure portal interface for a virtual machine named 'Breach-lab-vm'. The left sidebar shows the navigation menu with options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Monitor, Resource visualizer, Connect, Networking, Network settings, Load balancing, Application security groups, Network manager, Settings, and Availability + scale. The main content area is divided into two sections: 'Essentials' and 'Properties'. The 'Essentials' section provides a summary of the VM's configuration, including its resource group (Cryvion-Networks), status (Running), location (East Asia), subscription ID, operating system (Linux (ubuntu 24.04)), size (Standard B2s v2), primary NIC public IP (20.189.64.115), virtual network/subnet (Breach-lab-vm-vnet/default), DNS name (Not configured), health state (-), and time created (20/03/2026, 19:17 UTC). The 'Properties' section is further divided into 'Virtual machine' and 'Networking' tabs. The 'Virtual machine' tab shows details such as computer name (Breach-lab-vm), operating system (Linux (ubuntu 24.04)), VM generation (V2), and VM architecture (x64). The 'Networking' tab shows the public IP address (20.189.64.115) and the private IP address (10.0.0.4).

Explanation:

Ports configured:

Port Purpose

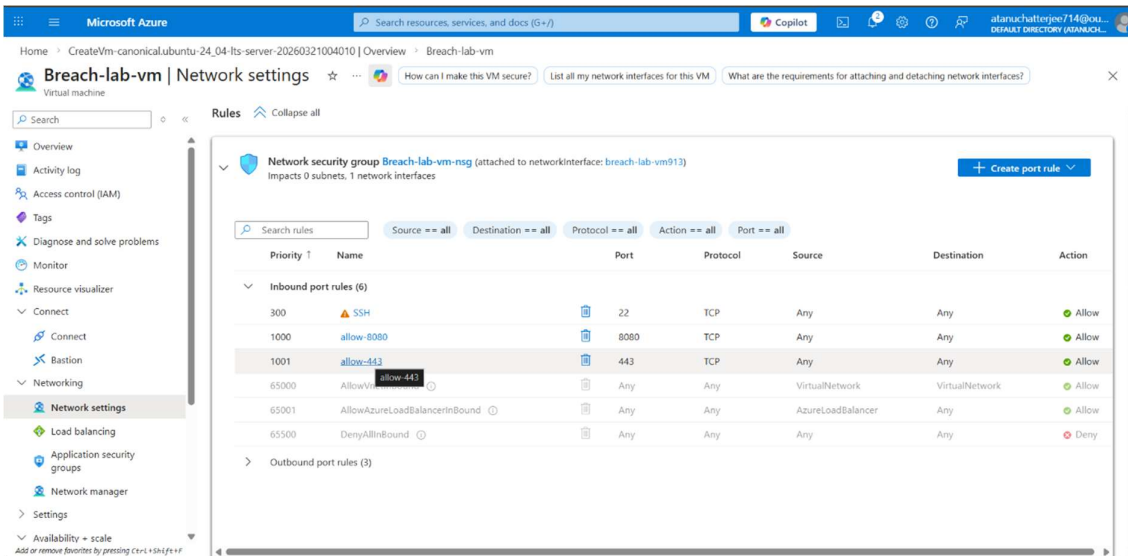
22 SSH

8080 DVWA

443 Wazuh

4. SSH Access to VM

Screenshot:

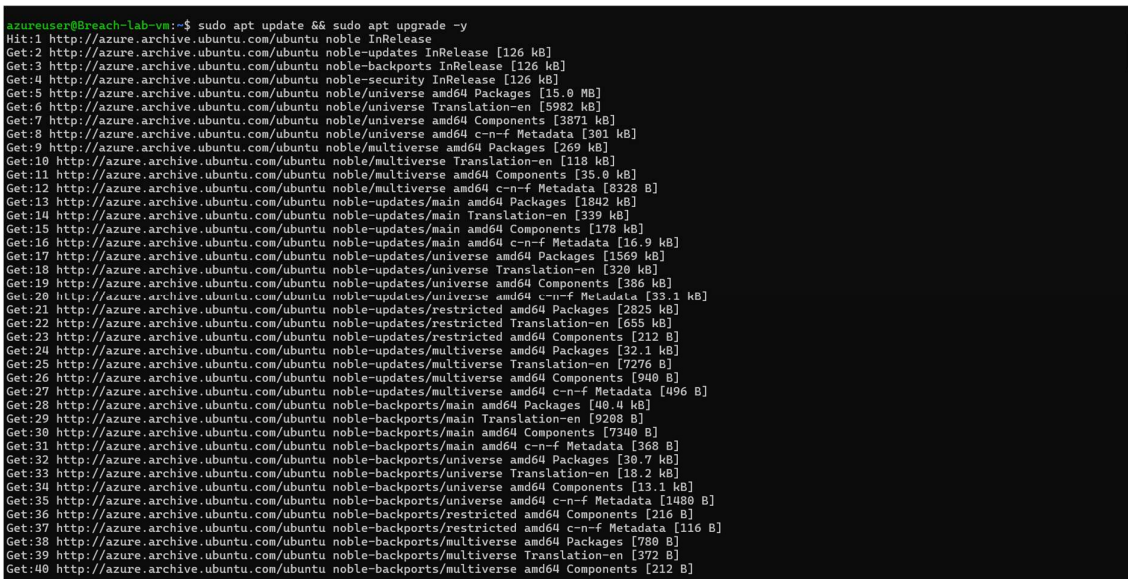


Explanation:

- SSH connection established successfully
- System ready for configuration

5. System Update

Screenshot:



Explanation:

- System packages updated

- Security patches applied

6. Docker Installation

Screenshot:

```
azureuser@reach-lab-vm:~$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1-24.04.1
azureuser@reach-lab-vm:~$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
Digest: sha256:85404b3c53951c3ff5d40de0972b1bb21fafa2e8daa235355baf44f33db9dbdd
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

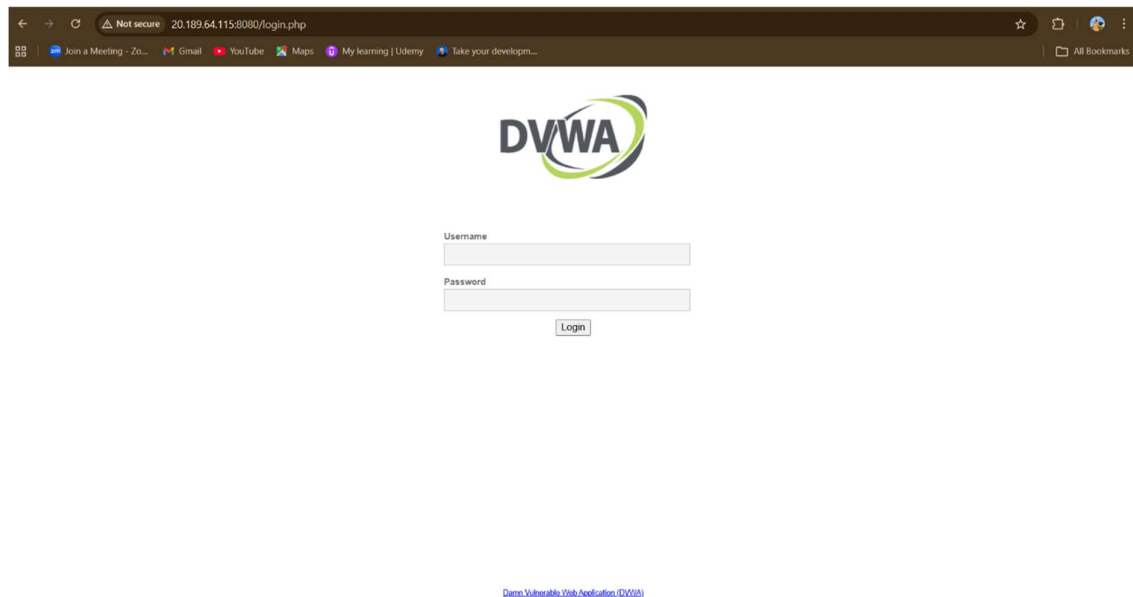
For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

Explanation:

- Docker installed
- Verified using hello-world container

7. DVWA Deployment

Screenshots:



Explanation:

- DVWA deployed using Docker

- Accessible via:

👉 <http://20.189.64.115:8080>

8. DVWA Setup

Explanation:

- Database configured
- Application ready for testing

9. Wazuh Installation

Screenshot:

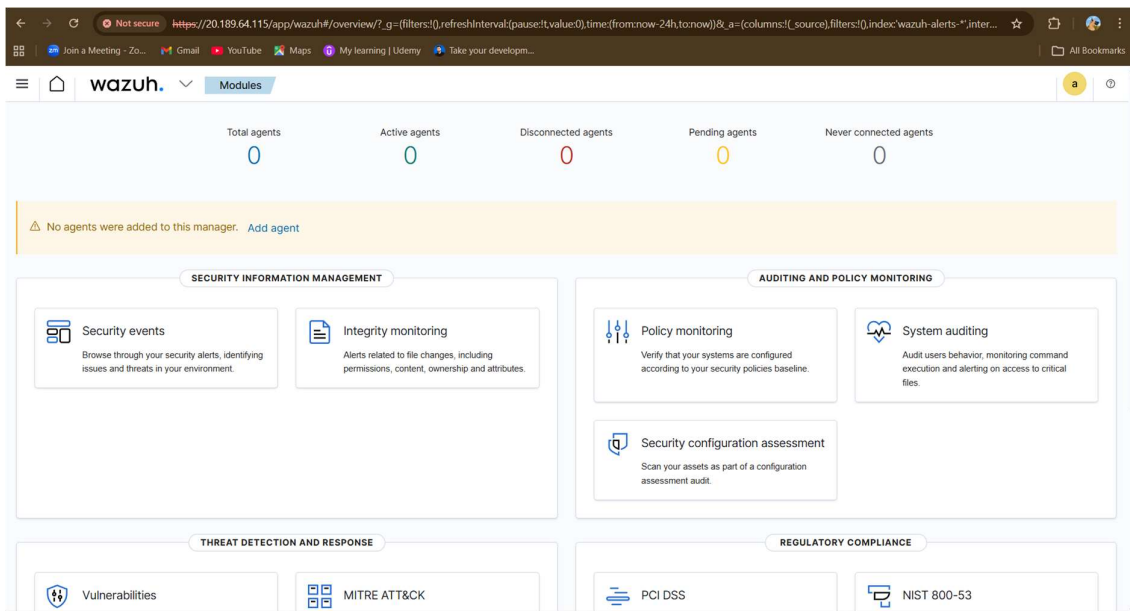
```
ireuseuser@reach-lab-vm:~$ sudo bash wazuh-install.sh -a -i
0/03/2026 19:39:46 INFO: Starting Wazuh installation assistant. Wazuh version: 4.7.5
0/03/2026 19:39:46 INFO: Verbose logging redirected to /var/log/wazuh-install.log
0/03/2026 19:39:53 WARNING: Hardware and system checks ignored.
0/03/2026 19:39:53 INFO: Wazuh web interface port will be 443.
0/03/2026 19:39:57 INFO: Wazuh repository added.
0/03/2026 19:39:57 INFO: --- Configuration files ---
0/03/2026 19:39:57 INFO: Generating configuration files.
0/03/2026 19:39:58 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passwords necessary for installation.
0/03/2026 19:39:59 INFO: --- Wazuh indexer ---
0/03/2026 19:39:59 INFO: Starting Wazuh indexer installation.
0/03/2026 19:40:45 INFO: Wazuh indexer installation finished.
0/03/2026 19:40:45 INFO: Wazuh indexer post-install configuration finished.
0/03/2026 19:40:45 INFO: Starting service wazuh-indexer.
0/03/2026 19:40:58 INFO: wazuh-indexer service started.
0/03/2026 19:40:58 INFO: Initializing Wazuh indexer cluster security settings.
0/03/2026 19:41:10 INFO: Wazuh indexer cluster initialized.
0/03/2026 19:41:10 INFO: --- Wazuh server ---
0/03/2026 19:41:10 INFO: Starting the Wazuh manager installation.
0/03/2026 19:41:52 INFO: Wazuh manager installation finished.
0/03/2026 19:41:52 INFO: Starting service wazuh-manager.
0/03/2026 19:42:03 INFO: wazuh-manager service started.
0/03/2026 19:42:03 INFO: Starting Filebeat installation.
0/03/2026 19:42:09 INFO: Filebeat installation finished.
0/03/2026 19:42:10 INFO: Filebeat post-install configuration finished.
0/03/2026 19:42:10 INFO: Starting service filebeat.
0/03/2026 19:42:11 INFO: filebeat service started.
0/03/2026 19:42:11 INFO: --- Wazuh dashboard ---
0/03/2026 19:42:11 INFO: Starting Wazuh dashboard installation.
0/03/2026 19:43:10 INFO: Wazuh dashboard installation finished.
0/03/2026 19:43:10 INFO: Wazuh dashboard post-install configuration finished.
0/03/2026 19:43:10 INFO: Starting service wazuh-dashboard.
0/03/2026 19:43:10 INFO: wazuh-dashboard service started.
0/03/2026 19:43:36 INFO: Initializing Wazuh dashboard web application.
0/03/2026 19:43:37 INFO: Wazuh dashboard web application initialized.
0/03/2026 19:43:37 INFO: --- Summary ---
0/03/2026 19:43:37 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
User: admin
Password: aGt?+TpEjs0FXt+1GKVa*1P3IrtwUvif
0/03/2026 19:43:37 INFO: Installation finished.
```

Explanation:

- Wazuh installed successfully
- Dashboard configured on port 443

10. Wazuh Dashboard

Screenshot:



Explanation:

- SIEM dashboard active
- Monitoring enabled

11. Attack Simulation

◆ 11.1 Network Scanning (Recon)

Screenshot:

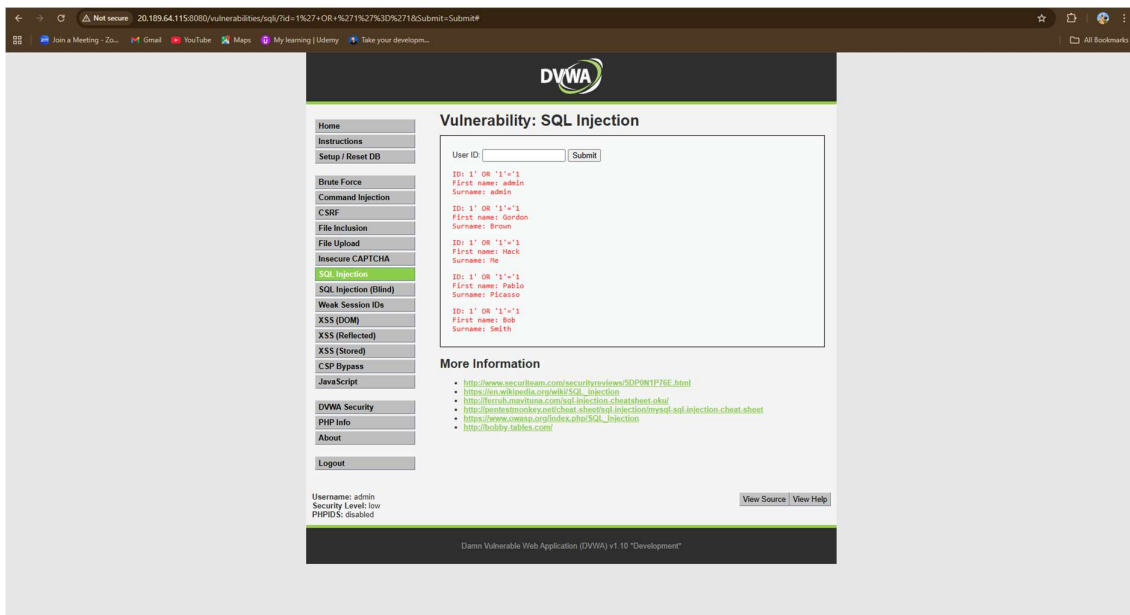
```
azureuser@breach-lab-vm:~$ sudo nmap -sS localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-20 20:09 UTC
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000040s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
443/tcp   open  https
8080/tcp   open  http-proxy
9200/tcp   open  wap-wsp
Nmap done: 1 IP address (1 host up) scanned in 0.21 seconds
```

Explanation:

- Attacker scanned system
- Open ports identified

◆ 11.2 SQL Injection Attack

Add Screenshot:



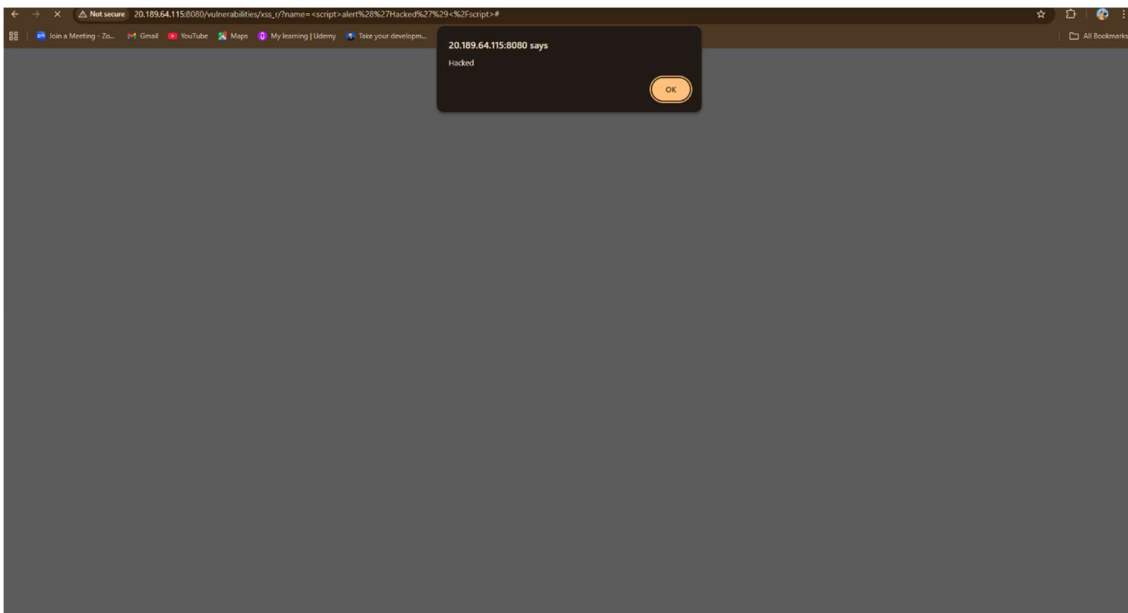
👉 (DVWA SQL injection result page)

📄 Explanation:

- Input: ' OR 1=1
- Result: Login bypass
- Database exposed

◆ 11.3 XSS Attack**

Screenshot:



Explanation:

- Script injected
 - Browser executed malicious code
-

12. Detection using Wazuh

 Pdf: C:\Users\atanu_3ne3sy2\Downloads\wazuh-module-overview-general-1774041037.pdf

Alerts Observed:

- PAM Login session
 - sudo to ROOT
 - dpkg installation
 - anomaly detection
-

13. Attack Mapping (MITRE ATT&CK)

Stage	Attack
Recon	Nmap

Stage	Attack
Initial Access	Login attempts
Execution	SQL Injection
Privilege Escalation	sudo
Persistence	system changes
Detection	Wazuh

14. Incident Response

- Suspicious activity detected
 - Logs analyzed
 - Unauthorized actions identified
 - System secured
-

15. Security Hardening

- Strong passwords
 - Firewall rules
 - Monitoring enabled
 - Zero Trust model
-

16. Results

- ✓ Successful attack simulation
 - ✓ Real-time detection using Wazuh
 - ✓ Logs verified attack behavior
-

17. Conclusion

This project successfully demonstrates a real-world enterprise breach scenario using containerized environments and SIEM detection tools. It highlights the importance of monitoring, detection, and security hardening.

